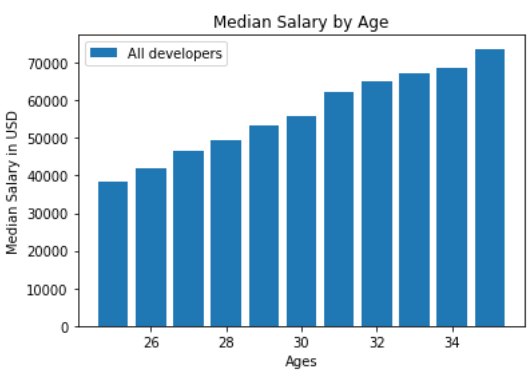
```
import pandas as pd
import matplotlib.pyplot as plt
```

```
x = [25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35]
```

```
devs_y = [38496, 42000, 46752, 49320, 53200, 56000, 62316, 64928, 67317, 68748, 73752]
```

1. Plotting the bar plot

```
plt.bar(x, devs_y, label="All developers")
plt.xlabel("Ages")
plt.ylabel("Median Salary in USD")
plt.title("Median Salary by Age")
plt.legend()
plt.show()
```

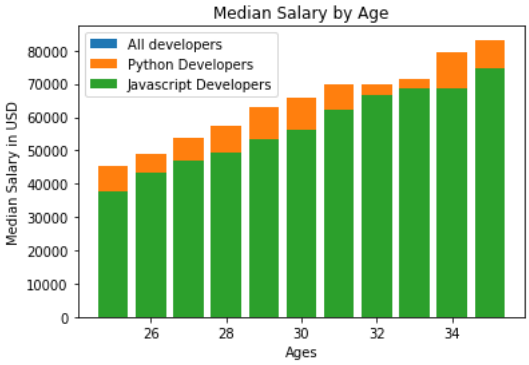


2. Adding more bars to the same plot

```
py_devs_y = [45372, 48876, 53850, 57287, 63016, 65998, 70003, 70000, 71418, 79674, 83238]
```

```
js_devs_y = [37810, 43515, 46823, 49293, 53437, 56373, 62375, 66674, 68745, 68746, 74583]
```

```
plt.bar(x, devs_y, label="All developers")
plt.bar(x, py_devs_y, label="Python Developers")
plt.bar(x, js_devs_y, label="Javascript Developers")
plt.xlabel("Ages")
plt.ylabel("Median Salary in USD")
plt.title("Median Salary by Age")
plt.legend()
plt.show()
```



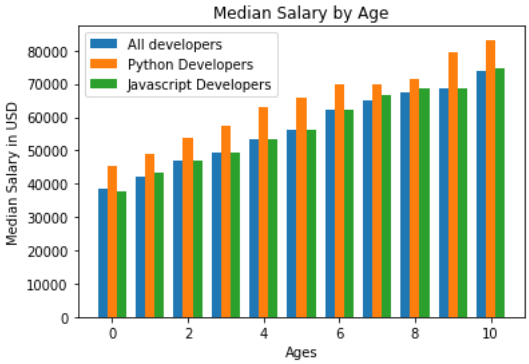
3. Adjusting the width of the plot

```
import numpy as np
```

```
x_indexes = np.arange(len(x))
```

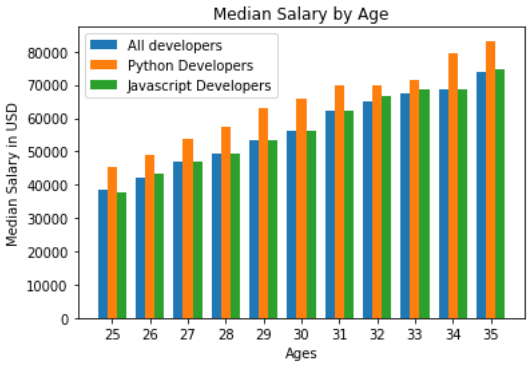
```
width = 0.25
```

```
plt.bar(x_indexes - width, devs_y, width=width, label="All developers")
plt.bar(x_indexes, py_devs_y, width = width, label="Python Developers")
plt.bar(x_indexes + width, js_devs_y, width=width, label="Javascript Developers")
plt.xlabel("Ages")
plt.ylabel("Median Salary in USD")
plt.title("Median Salary by Age")
plt.legend()
plt.show()
```



4. Changing the xlabels

```
plt.bar(x_indexes - width, devs_y, width=width, label="All developers")
plt.bar(x_indexes, py_devs_y, width = width, label="Python Developers")
plt.bar(x_indexes + width, js_devs_y, width=width, label="Javascript Developers")
plt.xlabel("Ages")
plt.ylabel("Median Salary in USD")
plt.title("Median Salary by Age")
plt.xticks(ticks=x_indexes, labels=x) #changing the xlabel
plt.legend()
plt.show()
```



5. Plotting the bar plot from pandas dataframe

```
import pandas as pd

from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
data = pd.read_csv('/content/drive/My Drive/data/data.csv')
```

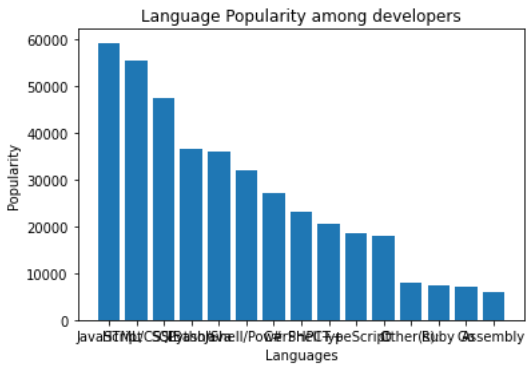
```
data.head()
```

	Responder_id	LanguagesWorkedWith
0	1	HTML/CSS;Java;JavaScript;Python
1	2	C++;HTML/CSS;Python
2	3	HTML/CSS
3	4	C;C++;C#;Python;SQL
4	5	C++;HTML/CSS;Java;JavaScript;Python;SQL;VBA

```
from collections import Counter


ids = data['Responder_id']
language_responses = data['LanguagesWorkedWith']


language_counter = Counter()


for response in language_responses:
    language_counter.update(response.split(";"))


languages = []
popularity = []

for item in language_counter.most_common(15):
    languages.append(item[0])
    popularity.append(item[1])


print(languages)
print(popularity)
```

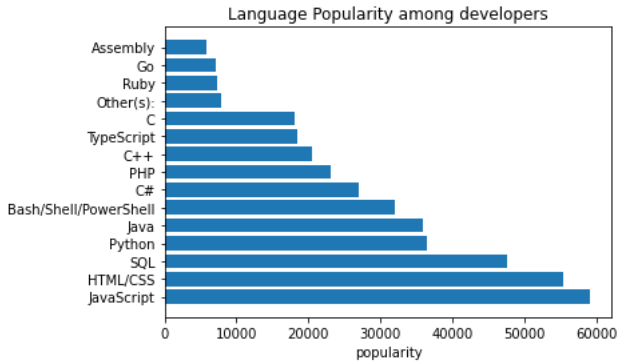
```
['JavaScript', 'HTML/CSS', 'SQL', 'Python', 'Java', 'Bash/Shell/PowerShell', 'C#', 'PHP', 'C++', 'TypeScript', 'C', 'Other(s):', 'Ru
[59219, 55466, 47544, 36443, 35917, 31991, 27097, 23030, 20524, 18523, 18017, 7920, 7331, 7201, 5833]
```

```
plt.bar(languages, popularity)
plt.xlabel("Languages")
plt.ylabel("Popularity")
plt.title("Language Popularity among developers")
plt.show()
```



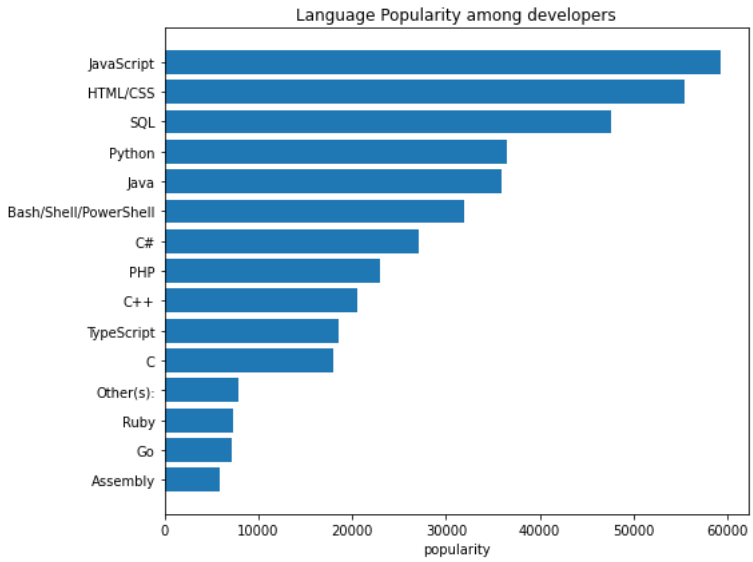
6. Plotting Horizontal bar chart

```
plt.barh(languages, popularity)
plt.xlabel("popularity")
plt.title("Language Popularity among developers")
plt.show()
```




```
languages.reverse()
popularity.reverse()
```

```
# https://insights.stackoverflow.com/survey/2020#technology-programming-scripting-and-markup-languages

plt.figure(figsize=(8,6))
plt.barh(languages, popularity)
plt.xlabel("popularity")
plt.title("Language Popularity among developers")
plt.tight_layout()
plt.show()
```



Show Your Creativity

Automobile Land Speed Records (GR 5-10)

In the first recorded automobile race in 1898, Count Gaston de Chasseloup-Laubat of Paris, France, drove 1 kilometer in 57 seconds for an average speed of 39.2 miles per hour(mph) or 63.1 kilometers per hour (kph). In 1904, Henry Ford drove his Ford Arrow across frozen Lake St. Clair, MI, at an average speed of 91.4 mph. Now, the North American Eagle is trying to break a land speed record of 800 mph. The Federation International deL'Automobile (FIA), the world's governing body for motor sport and land speed records,recorded the following land speed records.

```
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import pandas as pd
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
data = pd.read_csv('/content/drive/My Drive/data/LandRecords.csv')
```

```
data.head()
```

	Speed (mph)	Driver	Car	Engine	Date
0	407.447	Craig Breedlove	Spirit of America	GE J47	8/5/1963
1	413.199	Tom Green	Wingfoot Express	WE J46	10/2/1964
2	434.220	Art Arfons	Green Monster	GE J79	10/5/1964
3	468.719	Craig Breedlove	Spirit of America	GE J79	10/13/1964
4	526.277	Craig Breedlove	Spirit of America	GE J79	10/15/1965